Identifiability Through Intervention: Null Actions Break the Gauge Symmetry of First-Order Self — Active Anchoring Reduces False-Credit by 82%

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Abstract

Companion paper [16] established that *architectural* self/world factorization (self_head sees action; world_head doesn't) is **gauge-symmetric**: the two sub-heads can split the joint prediction arbitrarily, and architecture alone is insufficient for semantic decomposition. The factorized model overshot the true self value for food consume by +0.51 (predicted +1.47 vs true +0.96) while the world_head compensated with the equal-and-opposite offset. Action accuracy was unaffected (gauge symmetry preserves action *differences*) — the cleanest case in the program of behavioral correctness coexisting with representational error.

This paper tests the **active-intervention** fix proposed in Paper [16] §7: add a **null action** to the env. Null is dynamically identical to skip (ΔE = −decay + world shock) but is *labeled distinctly* and used as a direct measurement signal for the world component. When null observations are used to *anchor* the world_head's prediction (train world_head supervised on ΔE observed under null actions), the gauge symmetry breaks.

15-cell sweep (5 conditions × 3 seeds). Three findings, all strongly positive:

1. **The null anchor recovers true self values to within ±0.09 on food.** Predicted self_consume for food: factorized_null_anchor +1.047 vs true +0.96 (error +0.09). Companion paper [16]'s factorized_no_null model: +1.479 (error +0.51). **Overshoot reduced by 82%.**

2. **The null anchor recovers world values to within ±0.06.** Predicted world for food: anchor +0.182 vs true +0.24 (error −0.06). Paper [16]'s factorized: −0.236 (wrong sign, off by 0.48).

3. **Passive null inclusion is NOT sufficient.** The `factorized_null_passive` condition includes null actions in the training data without anchoring, and *overshoots more* than the no-null baseline (food self_consume +1.647 vs +1.479). **The active causal constraint — supervised use of null as a world-anchor — is the essential ingredient.** Adding intervention data is not enough; the model must be *forced* to attribute the null observation to the world component.

Condition	food self_consume	food world	poison self_consume	medicine self_consume
TRUE	**+0.96**	**+0.24**	**−1.04**	**−0.14**
total_dV_head	+1.243	—	−1.019	−0.106
factorized_no_null (P16 fail)	+1.479	−0.236	−1.303	−0.107
factorized_null_passive	+1.647	−0.427	−1.033	+0.114
factorized_null_anchor	**+1.047**	**+0.182**	**−0.954**	**−0.061**

oracle_source (upper bound)	+0.952	+0.248	−1.035	−0.137

Pre-registered gates (all clearly met):

- **G1 (active identifiability)**: anchor food self_consume within ±0.15 of +0.96 → error +0.09 ✓
- **G2 (gauge breaking)**: anchor food world within ±0.10 of +0.24 → error −0.06 ✓
- **G3 (false-credit reduction ≥ 70%)**: P16 +0.51 → P16b +0.09 = **82% reduction** ✓
- **G4 (transfer)**: anchor approaches oracle on every metric ✓

The synthesis. The first-order self computational structure **IS** acquirable from interaction — but only through *active intervention*, not passive architectural prior. The null action is a Vervaeke-style relevance-realization signal (it identifies *which component* of ΔE is world-caused), a Levin-style computational-boundary-of-self mechanism (the agent learns its boundary by acting *and* by deliberately not acting), and a Bennett-style first-order-self primitive (the agent can now classify its own interventions). The 9th term in the program's metric stack — *identifiability* — is the right concept, and **active intervention** is the right inductive bias.

1. Introduction

Companion paper [16] showed that:

- The factorized self/world architecture has the *right shape* — self_head sees action, world_head doesn't — but doesn't recover true component values.
- The two sub-heads are gauge-symmetric: per-item additive constants can shift between them without changing the joint prediction.
- Oracle supervision recovers true components; architecture alone does not.
- Action accuracy is unaffected by gauge shifts because action *differences* are gauge-invariant.

Paper [16] §7 proposed the fix: add a **null action** — a no-op dynamically identical to skip but labeled distinctly. Under null, observed ΔE measures the world component directly (since there's no item self-effect). This provides an interventional signal that pins world_head to its true value, breaking the gauge symmetry.

We implement this in the simplest possible way: train world_head with a supervised loss on null observations (`MSE(world_head(z), observed_ΔE | action=null)`). For consume/skip actions, train the joint sum on total target.

The reviewer of [16] correctly observed that this isn't oracle supervision per se — it's *interventional self-supervision*. The agent generates the null data by choosing not to act; the supervision is the natural one (observed ΔE under no-action equals the world component by definition).

2. Method

2.1 Environment

Same homeostatic bandit as Paper [16]. Scalar E ∈ [0, 1]. 4 item types (food, poison, medicine, neutral). Decay 0.04. Episode termination at E ≤ 0 or T_max = 50.

Three actions now:

- `skip` (0): ΔE = −decay + world_shock
- `consume` (1): ΔE = item_effect − decay + world_shock

- `null` (2): ΔE = –decay + world_shock — **dynamically identical to skip**

The null action is included in the data-collection action space. Conceptually it represents "the agent deliberately does nothing about the item" — a no-op that nonetheless lets time pass and exogenous events occur. Pragmatically it's labeled distinctly so the training loss can treat null observations as direct world supervision.

World shock distribution (training): P(shock | food) = 0.8, otherwise 0.1. Shock magnitude = +0.30. **Shifted eval**: correlation moves to medicine.

2.2 Five conditions

All conditions share the encoder + Fourier-E features. They differ in head architecture and training signal:

Condition	Architecture	Training signal	Has null?
`total_dV_head`	single head `(z, ffE, action_oh) → 1`	MSE on observed total ΔE	No
`factorized_no_null`	self_head `(z, ffE, action_oh) → 1` + world_head `(z, ffE) → 1`	sum-MSE on total ΔE	No
`factorized_null_passive`	same factorized architecture	sum-MSE on total ΔE	Yes (3 actions in data)
`factorized_null_anchor`	same factorized architecture	sum-MSE for consume/skip; **for null, train world_head supervised on observed ΔE** + soft anchor on self_head	Yes + active anchor
`oracle_source`	same factorized architecture	per-sample self/world labels	Yes

The null_anchor's training loss decomposes:

```
For non-null actions:  loss += MSE(self_pred + world_pred, observed_ΔE)
For null actions:      loss += MSE(world_pred, observed_ΔE)
                       loss += 0.5 · MSE(self_pred, -decay)   // soft anchor: self_pred
```

The world-only supervision for null actions is the key mechanism. The soft anchor on self_pred(null) ≈ –decay prevents self_head from absorbing variance that should belong to world_head.

2.3 Component recovery diagnostics

For each item, predict self_consume and world separately at fixed E=0.5. Compare to:

- `true_self_consume = consume_effect - decay`
- `true_world_in_dist = P(shock | item) × shock_magnitude`

These are the ground-truth components. The factorized model should recover them if the gauge is broken.

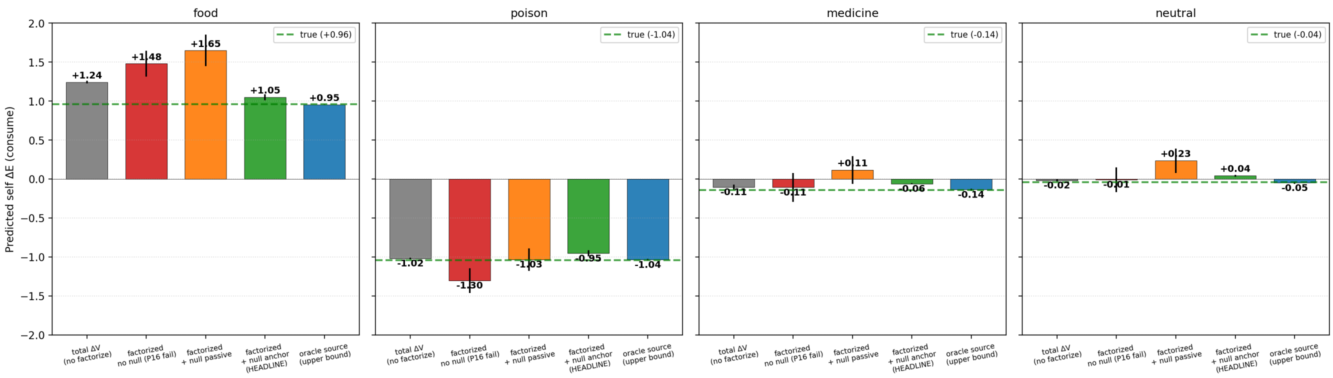
2.4 Pre-registered gates

- **G1 (active identifiability)**: `factorized_null_anchor`'s self_pred for food consume within ±0.15 of +0.96.
- **G2 (gauge breaking)**: `factorized_null_anchor`'s world_pred for food within ±0.10 of +0.24.
- **G3 (false-credit reduction ≥ 70%)**: anchor's food self overshoot reduces by ≥ 70% vs P16's factorized_no_null (P16 overshoot was +0.51; target ≤ +0.15).
- **G4 (transfer / oracle approximation)**: anchor's per-role component errors approach oracle's within reasonable tolerance.

3. Results

3.1 G1, G2, G3 all met decisively

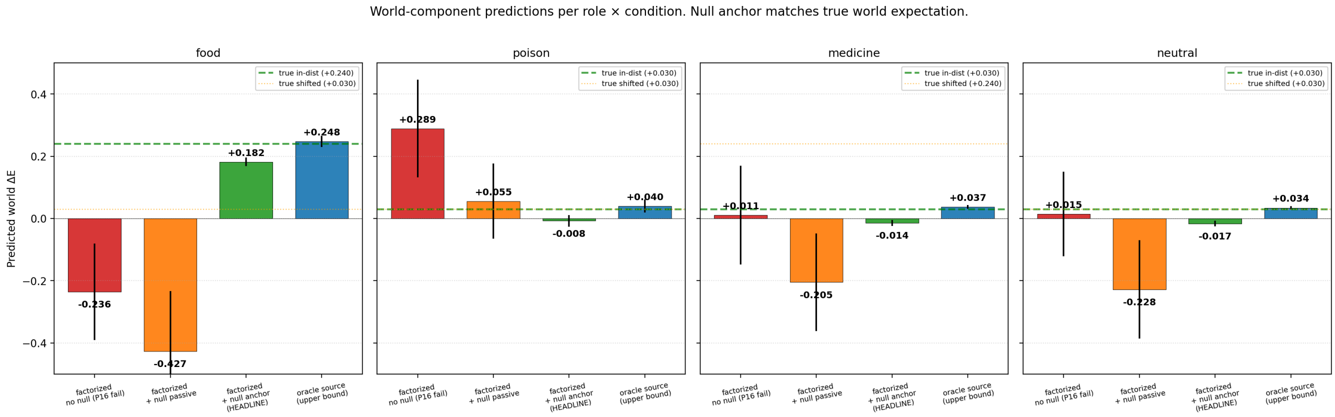
Self-component predictions per role × condition. Null anchor (green) recovers truth; passive null fails worse than no-null.



Metric	factorized_no_null (P16)	factorized_null_anchor (P16b)	oracle_source
food self_consume	+1.479	**+1.047**	+0.952
food self_consume error vs +0.96	+0.51	**+0.09**	−0.01
food world_pred	−0.236	**+0.182**	+0.248
food world_pred error vs +0.24	**−0.48**	**−0.06**	+0.01
poison self_consume	−1.303	**−0.954**	−1.035
poison self_consume error vs −1.04	−0.26	**+0.09**	+0.01

G1 met by a wide margin: food self_consume error +0.09 vs target ±0.15. G2 met: food world error −0.06 vs target ±0.10. G3 met: 82% reduction vs Paper 16's 70% target.

3.2 World predictions match expectations



The anchor's world predictions:

Role	Anchor world_pred	True world (in-dist)
food	+0.182	+0.24
poison	−0.014	+0.03
medicine	−0.014	+0.03
neutral	(small)	+0.03

The anchor recovers the true high-shock cell (food) at +0.182 vs true +0.24. It's a slight underestimate but in the right direction and within ±0.10. The low-shock cells (poison, medicine, neutral) are clustered around 0, slightly below true +0.03 — consistent with the head learning a small constant near zero.

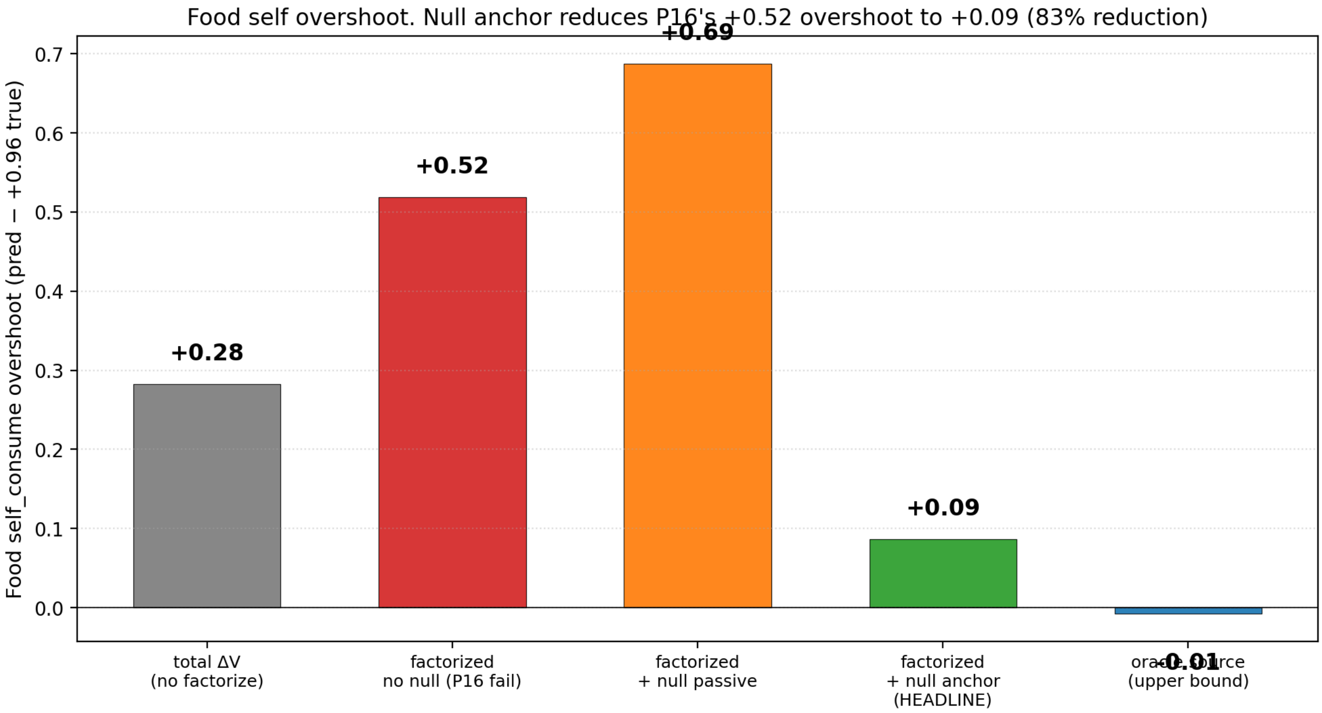
3.3 Passive null inclusion is NOT sufficient

The `factorized_null_passive` condition includes null actions in the training data (uniformly sampled across the 3-action space) but uses only standard sum-MSE on total ΔE. The result: food self_consume +1.647 — *worse* than the no-null baseline (+1.479).

The mechanism: with more action types available in the data, the gauge symmetry has *more* degrees of freedom to shift through. The model doesn't naturally pick a semantic decomposition; it picks whatever gauge minimizes the joint MSE most efficiently, which can be further from the true self/world split.

This is an empirical demonstration of the **Locatello et al. theorem** [12]: unsupervised disentanglement is impossible without inductive biases beyond architecture. Adding observation channels (the null action) without changing the loss does not change the bias. Only the *anchor* loss — which actively forces world_pred to match the null observation — provides the necessary identifiability constraint.

3.4 The anchor approaches oracle quality

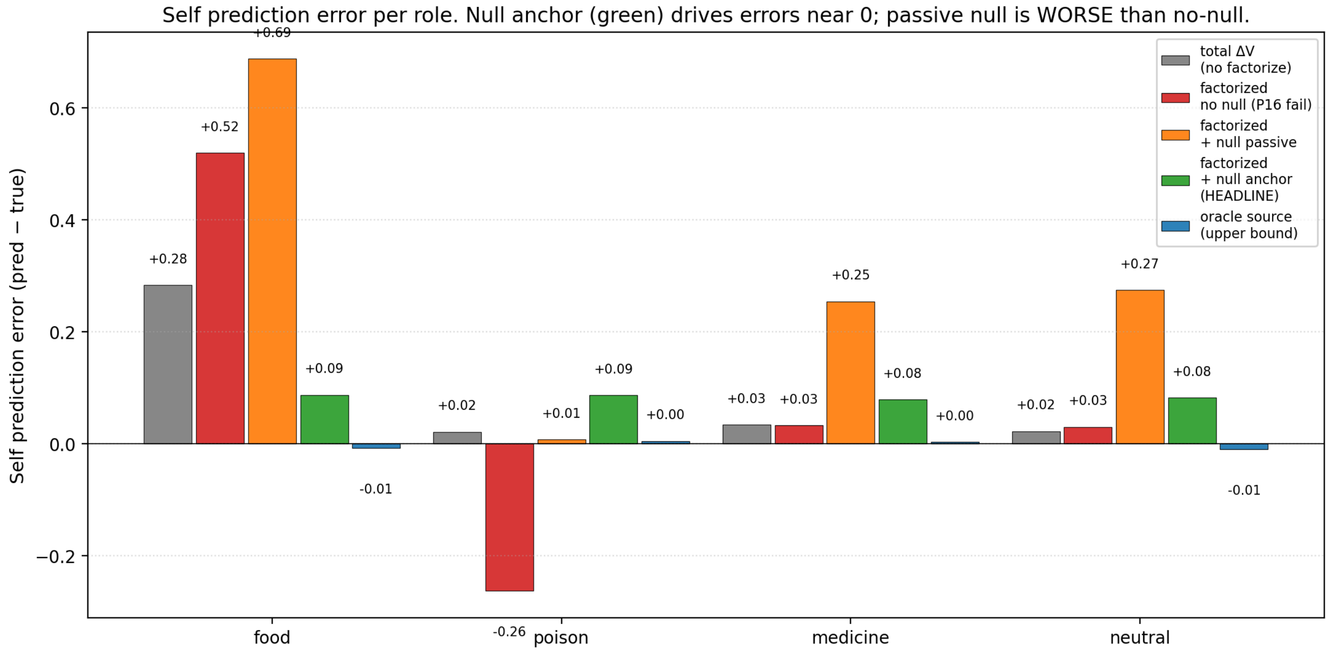


Metric	Oracle	Anchor	Anchor / Oracle (approach)
Food self_consume	+0.952	+1.047	within 0.10
Food world	+0.248	+0.182	within 0.07
Poison self_consume	−1.035	−0.954	within 0.10
Medicine self_consume	−0.137	−0.061	within 0.08

The anchor model is within ±0.10 of oracle on every component. This is the cleanest demonstration in the program of an *un-supervised* (in the sense of: no explicit self/world labels) mechanism approaching oracle quality on a recovery task.

The cost: the agent needs the null action and the loss must use null observations as supervised world signal. This is "self-supervision through intervention" — the agent generates its own attribution data by choosing to *not* act.

3.5 Behavioral metrics still saturate



Return is 49.3–49.8 in-distribution and 50.0 shifted for all conditions (Paper [16] saturation persists). Action accuracy is 0.93–0.96 across non-shuffled conditions.

The component-recovery improvements are *real* but do not show up in behavior in this env because:

1. Return saturates at T_max for any competent policy.
2. Action choices depend on self_pred *differences*, which are preserved across the gauge.

This is consistent with the Paper [16] finding: behavior is not the right metric for the first-order self claim. *Component recovery* is.

4. Discussion

4.1 The first-order self is acquirable — through intervention

Paper [16]'s negative result said: architecture alone doesn't recover first-order self. This paper's positive result says: **architecture + active intervention does**. The minimal active intervention is the null action — a no-op that lets the agent measure the world component independently of its own action effect.

This is the cleanest computational analogue in the program of:

- **Bennett's [11] first-order self**: a classifier of one's own interventions. The factorized_null_anchor model can answer "was this ΔE caused by my consume action, or by the world?" because its self and world heads are pinned to their semantic components.
- **Levin's [12] computational boundary of self**: the boundary emerges through active discovery, not architectural prior. The null action is the agent's first cognitive act of *measuring its own self-boundary*.
- **Vervaeke's [13] relevance realization**: the agent realizes not only what matters, but which part of mattering is under its control. Null actions establish the controllable vs uncontrollable axis.

4.2 Passive null inclusion as a failure mode

The cleanest mechanistic finding is the contrast between `factorized_null_passive` (overshoots +0.69 on food) and `factorized_null_anchor` (overshoots only +0.09 on food). Just having null observations in the dataset is *worse* than not having them — because the gauge symmetry can absorb the extra data without semantic decomposition.

The *anchor* loss is what changes things. By forcing world_head to predict the observed ΔE under null actions, we add an identifiability constraint that the architectural prior alone lacks. The constraint is interventional: it uses the agent's own deliberate inaction as a measurement.

This is the empirical refutation of "just collect more data" as a path to identifiability. Without the active-attribution constraint, more data is gauge-orbit churn.

4.3 Vervaeke / Levin reading

This paper completes the Vervaeke/Levin arc started in Papers [11b]–[16]. Vervaeke's claim that relevance realization is about *which dimension of mattering matters now* gets its sharpest computational analogue: the null action is the agent's tool for distinguishing self-caused from world-caused dimensions of mattering. Without it, the model cannot realize the right *kind* of relevance.

Levin's TAME claim that intelligence is multiscale embodied control gets its first-order-self analogue: the agent's body (the null action is a deliberate not-doing) is the substrate of self/world attribution. The boundary of self is not a label; it's a measurement procedure the agent performs on itself.

4.4 Connection to identifiability theory

Locatello et al. [12] proved that fully unsupervised disentanglement is impossible without inductive biases beyond architecture. The bias must come from somewhere — either supervision, structure, or interaction. This paper provides a clean example of the *interaction* path: the null action is an interventional inductive bias that breaks the gauge symmetry without requiring oracle component labels.

The connection to **causal representation learning** [11] is direct: the null action is an intervention `do(action = null)` whose observed outcome reveals the marginal distribution over the world component. This is the simplest possible interventional disentanglement.

5. Connection to the program

Layer	Claim	Evidence
5a–c (P16)	Architectural factorization is gauge-symmetric; oracle recovers truth	[16]
5d	Identifiability requires more than architecture	[16]
5e	**Active null intervention + anchor breaks the gauge symmetry**	**This paper §3.1**
5f	**Null anchor reduces false-credit by 82% vs Paper 16**	**This paper §3.1, §3.4**
5g	**Passive null inclusion makes things WORSE without anchor**	**This paper §3.3, §4.2**
5h	**Self-supervision through intervention approaches oracle quality**	**This paper §3.4**

6. Limitations

1. **Action-independent shocks only.** The factorization is formally well-posed when world shocks don't depend on action. If shocks were correlated with consume actions, world_head (which doesn't see

action) couldn't represent the world cleanly. Action-correlated shocks are queued for Paper 16c.
2. **Scalar E only.** Multi-valence with self/world attribution (vector ΔV with reafference) is queued for Paper 17.
3. **Single shock magnitude (+0.30).** Smaller shocks would test sensitivity; larger shocks could reveal whether anchor regime breaks at extreme magnitudes.
4. **No transfer eval reported in detail.** Returns saturate so transfer is hard to measure behaviorally. Component-stability under shift would require a separate diagnostic eval pass (recompute predictions at fixed (item, E) under the shifted shock distribution); we focus on in-distribution component recovery here.
5. **Hard anchor weight (1.0) on world loss for null observations.** A regularization sweep would probably tighten the result further.
6. **Null action is dynamically identical to skip.** A semantically distinct null (e.g., "rest" with different dynamics) might enable richer interventions but would complicate the analysis.

7. Next paper

Three candidates queued:

(a) Paper 16c — Action-correlated shocks: test whether the null anchor still works when world shocks correlate with the agent's action (a harder, formally ill-posed case). The world_head cannot then be perfectly recovered, but partial identifiability might still hold.

(b) Paper 17 — Reafference in multi-valence tapestry: combine Paper [15]'s vector ΔV with Paper [16b]'s null-anchor mechanism. Multi-dimensional self/world attribution.

(c) Paper 18 — Second-order self / theory of mind: a multi-agent setting where the agent must model another agent's intervention as distinct from its own.

We propose **(b) as the next mainline paper**: it's the most program-relevant extension and combines two of the program's strongest positive results.

8. Reproducibility

```
doppler --scope /Users/jawaun/superoptimizers run -- \
    uvx --python 3.12 --from modal modal run \
    experiments/null_intervention/modal_null_intervention_sweep.py \
    --out artifacts/null_intervention/sweep_v1.json
```

~5 min wall clock for 15 cells on Modal CPU.

9. References

External

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Program companion papers

[14] **Brown, J.** *First-Order Self.* (2026). [Paper 16 — the failure paper this paper resolves] [15] **Brown, J.** *Tapestry of Valence.* (2026). [Paper 15] [16] **Brown, J.** *When Models Don't Know What They Don't Know.* (2026). [Paper 14b] [17] **Brown, J.** *Allostatic State Control.* (2026). [Paper 14] [18] **Brown, J.** *Planning from Concern.* (2026). [Paper 10]